

# KIT SAFETY DATA SHEET



**Kit Product Name** Coomassie Stain Kit

**Kit Catalogue Number(s)** 1610435, 1610435EDU

**Revision date** 31-Aug-2022

## Kit Contents

| Catalogue Number(s)                      | Product Name                                     |
|------------------------------------------|--------------------------------------------------|
| 1610436, 1610437, 1610436EDU, 1610437EDU | Coomassie Brilliant Blue R-250 Staining Solution |
| 1610438, 1610439, 1610438EDU, 1610439EDU | Coomassie Brilliant Blue R-250 Destain Solution  |



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
Regulation (EC) No. 1907/2006 and Regulation (EC) No. 1272/2008

Revision date 31-Aug-2022

Revision Number 1.2

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### 1.1. Product identifier

**Product Name** Coomassie Brilliant Blue R-250 Staining Solution

**Catalogue Number(s)** 1610436, 1610437, 1610436EDU, 1610437EDU

**Pure substance/mixture** Mixture

Contains Methanol

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

**Recommended use** Laboratory chemicals

**Uses advised against** No information available

### 1.3. Details of the supplier of the safety data sheet

#### **Corporate Headquarters**

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### **Manufacturer**

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### **Legal Entity / Contact Address**

Bio-Rad Laboratories Ltd  
The Junction  
Station Road  
Watford, WD17 1ET  
UK

Bio-Rad Laboratories Pvt. Ltd.  
Bio-Rad House  
86-87, Udyog Vihar Phase IV Gurgaon  
122005  
Haryana India

Bio-Rad Laboratories (Pty) Ltd.  
34 Bolton Road  
Parkwood, Johannesburg 2193  
South Africa

For further information, please contact

#### **Technical Service**

00800 00246 723  
Ireland: Techsupport.UK@bio-rad.com  
India: support.india@bio-rad.com  
South Africa: cdg\_techsupport\_eemea@bio-rad.com

### 1.4. Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC Ireland: 353-19014670  
CHEMTREC India: 000-800-100-7141  
CHEMTREC South Africa: 0-800-983-611

## SECTION 2: Hazards identification

### 2.1. Classification of the substance or mixture

Regulation (EC) No 1272/2008

|                         |                     |
|-------------------------|---------------------|
| Acute toxicity - Oral   | Category 3 - (H301) |
| Acute toxicity - Dermal | Category 3 - (H311) |

|                                                  |                     |
|--------------------------------------------------|---------------------|
| Acute toxicity - Inhalation (Dusts/Mists)        | Category 4 - (H332) |
| Skin corrosion/irritation                        | Category 1 - (H314) |
| Serious eye damage/eye irritation                | Category 1 - (H318) |
| Specific target organ toxicity (single exposure) | Category 1          |
| Corrosive to metals                              | Category 1          |
| Flammable liquids                                | Category 3 - (H226) |

## 2.2. Label elements

Contains Methanol



Signal word  
Danger

### Hazard statements

H301 - Toxic if swallowed  
H311 - Toxic in contact with skin  
H314 - Causes severe skin burns and eye damage  
H332 - Harmful if inhaled  
H370 - Causes damage to organs  
H290 - May be corrosive to metals  
H226 - Flammable liquid and vapor

### Precautionary Statements - EU (§28, 1272/2008)

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water [or shower]  
P331 - Do NOT induce vomiting  
P370 + P378 - In case of fire: Use dry chemical, CO<sub>2</sub>, water spray or alcohol-resistant foam to extinguish  
P403 + P235 - Store in a well-ventilated place. Keep cool  
P301 + P330 + P331 - IF SWALLOWED: Rinse mouth. Do NOT induce vomiting  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P501 - Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

## 2.3. Other hazards

Harmful to aquatic life.

## SECTION 3: Composition/information on ingredients

### 3.1 Substances

Not applicable

### 3.2 Mixtures

| Chemical name       | Weight-% | REACH registration number | EC No     | Classification according to Regulation (EC) No. 1272/2008 [CLP]                                              | Specific concentration limit (SCL) | M-Factor | M-Factor (long-term) |
|---------------------|----------|---------------------------|-----------|--------------------------------------------------------------------------------------------------------------|------------------------------------|----------|----------------------|
| Methanol<br>67-56-1 | 20 - 35  | No data available         | 200-659-6 | Acute Tox. 3 (H301)<br>Acute Tox. 3 (H311)<br>Acute Tox. 3 (H331)<br>STOT SE 1 (H370)<br>Flam. Liq. 2 (H225) | STOT SE 1 ::<br>C>=1%              | -        | -                    |

|                        |         |                   |           |                                             |                                                                                                                             |   |   |
|------------------------|---------|-------------------|-----------|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---|---|
| Acetic acid<br>64-19-7 | 10 - 20 | No data available | 200-580-7 | Skin Corr. 1A (H314)<br>Flam. Liq. 3 (H226) | Eye Irrit. 2 ::<br>10%≤C<25%<br>Skin Corr. 1A ::<br>C≥90%<br>Skin Corr. 1B ::<br>25%≤C<90%<br>Skin Irrit. 2 ::<br>10%≤C<25% | - | - |
|------------------------|---------|-------------------|-----------|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---|---|

**Full text of H- and EUH-phrases: see section 16**

#### Acute Toxicity Estimate

If LD50/LC50 data is not available or does not correspond to the classification category, then the appropriate conversion value from CLP Annex I, Table 3.1.2, is used to calculate the acute toxicity estimate (ATEmix) for classifying a mixture based on its components

| Chemical name          | Oral LD50 mg/kg | Dermal LD50 mg/kg | Inhalation LC50 - 4 hour - dust/mist - mg/L | Inhalation LC50 - 4 hour - vapor - mg/L | Inhalation LC50 - 4 hour - gas - ppm |
|------------------------|-----------------|-------------------|---------------------------------------------|-----------------------------------------|--------------------------------------|
| Methanol<br>67-56-1    | 6200            | 15840             | No data available                           | 41.6976                                 | No data available                    |
| Acetic acid<br>64-19-7 | 3310            | 1060              | 11.4                                        | No data available                       | No data available                    |

This product does not contain candidate substances of very high concern at a concentration ≥0.1% (Regulation (EC) No. 1907/2006 (REACH), Article 59)

## SECTION 4: First aid measures

### 4.1. Description of first aid measures

|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b>                     | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Inhalation</b>                         | Remove to fresh air. If breathing has stopped, give artificial respiration. Get medical attention immediately. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. If breathing is difficult, (trained personnel should) give oxygen. Delayed pulmonary edema may occur. Get immediate medical advice/attention. |
| <b>Eye contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Keep eye wide open while rinsing. Do not rub affected area. Remove contact lenses, if present and easy to do. Continue rinsing. Get immediate medical advice/attention.                                                                                                                                                                                                               |
| <b>Skin contact</b>                       | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Get immediate medical advice/attention.                                                                                                                                                                                                                                                                                                                                  |
| <b>Ingestion</b>                          | Rinse mouth. Never give anything by mouth to an unconscious person. Do NOT induce vomiting. Get immediate medical advice/attention.                                                                                                                                                                                                                                                                                                                                            |
| <b>Self-protection of the first aider</b> | Remove all sources of ignition. Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. Use personal protective equipment as required. See section 8 for more information. Avoid contact with skin, eyes or clothing. Avoid direct contact with skin. Use barrier to give mouth-to-mouth resuscitation. Avoid breathing vapors or mists.                                              |

### 4.2. Most important symptoms and effects, both acute and delayed

|                 |                                                                        |
|-----------------|------------------------------------------------------------------------|
| <b>Symptoms</b> | Burning sensation. Coughing and/ or wheezing. Difficulty in breathing. |
|-----------------|------------------------------------------------------------------------|

### 4.3. Indication of any immediate medical attention and special treatment needed

|                           |                                                                                      |
|---------------------------|--------------------------------------------------------------------------------------|
| <b>Note to physicians</b> | Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. |
|---------------------------|--------------------------------------------------------------------------------------|

Possible perforation of stomach or esophagus should be investigated. Do not give chemical antidotes. Asphyxia from glottal edema may occur. Marked decrease in blood pressure may occur with moist rales, frothy sputum, and high pulse pressure.

## SECTION 5: Firefighting measures

### 5.1. Extinguishing media

**Suitable Extinguishing Media** Dry chemical. Carbon dioxide (CO<sub>2</sub>). Water spray. Alcohol resistant foam.

**Large Fire** CAUTION: Use of water spray when fighting fire may be inefficient.

**Unsuitable extinguishing media** Do not scatter spilled material with high pressure water streams.

### 5.2. Special hazards arising from the substance or mixture

**Specific hazards arising from the chemical** Risk of ignition. Keep product and empty container away from heat and sources of ignition. In the event of fire, cool tanks with water spray. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. The product causes burns of eyes, skin and mucous membranes. Thermal decomposition can lead to release of irritating gases and vapors.

### 5.3. Advice for firefighters

**Special protective equipment and precautions for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

## SECTION 6: Accidental release measures

### 6.1. Personal precautions, protective equipment and emergency procedures

**Personal precautions** Evacuate personnel to safe areas. Use personal protective equipment as required. See section 8 for more information. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Keep people away from and upwind of spill/leak. ELIMINATE all ignition sources (no smoking, flares, sparks or flames in immediate area). Pay attention to flashback. Take precautionary measures against static discharges. All equipment used when handling the product must be grounded. Do not touch or walk through spilled material. Attention! Corrosive material. Avoid breathing vapors or mists.

**Other information** Ventilate the area. Refer to protective measures listed in Sections 7 and 8.

**For emergency responders** Use personal protection recommended in Section 8.

### 6.2. Environmental precautions

**Environmental precautions** Refer to protective measures listed in Sections 7 and 8. Prevent further leakage or spillage if safe to do so. Prevent product from entering drains. Should not be released into the environment. Do not allow to enter into soil/subsoil.

### 6.3. Methods and material for containment and cleaning up

**Methods for containment** Stop leak if you can do it without risk. Do not touch or walk through spilled material. A vapor suppressing foam may be used to reduce vapors. Dike far ahead of spill to collect runoff water. Keep out of drains, sewers, ditches and waterways. Absorb with earth, sand or other non-combustible material and transfer to containers for later disposal.

**Methods for cleaning up** Take precautionary measures against static discharges. Dam up. Soak up with inert absorbent material. Pick up and transfer to properly labeled containers.

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

### 6.4. Reference to other sections

**Reference to other sections** See section 8 for more information. See section 13 for more information.

## SECTION 7: Handling and storage

### 7.1. Precautions for safe handling

#### Advice on safe handling

Use personal protection equipment. Avoid breathing vapors or mists. Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. Use grounding and bonding connection when transferring this material to prevent static discharge, fire or explosion. Use spark-proof tools and explosion-proof equipment. Keep in an area equipped with sprinklers. Use according to package label instructions. Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Take off contaminated clothing and wash before reuse. In case of insufficient ventilation, wear suitable respiratory equipment. Handle product only in closed system or provide appropriate exhaust ventilation. Do not eat, drink or smoke when using this product.

#### General hygiene considerations

Do not eat, drink or smoke when using this product. Contaminated work clothing should not be allowed out of the workplace. Regular cleaning of equipment, work area and clothing is recommended. Wash hands before breaks and immediately after handling the product. Wear suitable gloves and eye/face protection. Avoid contact with skin, eyes or clothing. Remove and wash contaminated clothing and gloves, including the inside, before re-use.

### 7.2. Conditions for safe storage, including any incompatibilities

#### Storage Conditions

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks, flame and other sources of ignition (i.e., pilot lights, electric motors and static electricity). Keep in properly labeled containers. Do not store near combustible materials. Keep in an area equipped with sprinklers. Store in accordance with the particular national regulations. Store in accordance with local regulations. Protect from moisture. Store locked up. Keep out of the reach of children. Store away from other materials. Store according to product and label instructions.

### 7.3. Specific end use(s)

**Risk Management Methods (RMM)** The information required is contained in this Safety Data Sheet.

## SECTION 8: Exposure controls/personal protection

### 8.1. Control parameters

#### Exposure Limits

| Chemical name          | European Union                                  | Austria                                                                                         | Belgium                                                                                         | Bulgaria                                                                                         | Croatia                                                                                            |
|------------------------|-------------------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Methanol<br>67-56-1    | TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup><br>* | TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup><br>STEL 800 ppm<br>STEL 1040 mg/m <sup>3</sup><br>H* | TWA: 200 ppm<br>TWA: 266 mg/m <sup>3</sup><br>STEL: 250 ppm<br>STEL: 333 mg/m <sup>3</sup><br>* | TWA: 200 ppm<br>TWA: 260.0 mg/m <sup>3</sup><br>K*                                               | TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup><br>*                                                    |
| Acetic acid<br>64-19-7 | -                                               | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL 20 ppm<br>STEL 50 mg/m <sup>3</sup>            | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 15 ppm<br>STEL: 38 mg/m <sup>3</sup>          | STEL: 50 mg/m <sup>3</sup><br>STEL: 20 ppm<br>TWA: 25 mg/m <sup>3</sup><br>TWA: 10 ppm           | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 20 ppm<br>STEL: 50 mg/m <sup>3</sup>             |
| Chemical name          | Cyprus                                          | Czech Republic                                                                                  | Denmark                                                                                         | Estonia                                                                                          | Finland                                                                                            |
| Methanol<br>67-56-1    | *<br>TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup> | TWA: 250 mg/m <sup>3</sup><br>Ceiling: 1000 mg/m <sup>3</sup><br>*                              | TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup><br>H*                                                | TWA: 200 ppm<br>TWA: 250 mg/m <sup>3</sup><br>STEL: 250 ppm<br>STEL: 350 mg/m <sup>3</sup><br>A* | TWA: 200 ppm<br>TWA: 270 mg/m <sup>3</sup><br>STEL: 250 ppm<br>STEL: 330 mg/m <sup>3</sup><br>iho* |
| Acetic acid<br>64-19-7 | STEL: 50 mg/m <sup>3</sup><br>STEL: 20 ppm      | TWA: 25 mg/m <sup>3</sup><br>Ceiling: 50 mg/m <sup>3</sup>                                      | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup>                                                        | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup>                                                         | TWA: 5 ppm<br>TWA: 13 mg/m <sup>3</sup>                                                            |

|                        |                                                                                                                     |                                                                                        |                                                                                                  |                                                                                                                                         |                                                                                                                                                                                                                                       |
|------------------------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup>                                                                            |                                                                                        |                                                                                                  | STEL: 10 ppm<br>STEL: 25 mg/m <sup>3</sup>                                                                                              | STEL: 10 ppm<br>STEL: 25 mg/m <sup>3</sup>                                                                                                                                                                                            |
| Chemical name          | France                                                                                                              | Germany                                                                                | Germany MAK                                                                                      | Greece                                                                                                                                  | Hungary                                                                                                                                                                                                                               |
| Methanol<br>67-56-1    | TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup><br>STEL: 1000 ppm<br>STEL: 1300 mg/m <sup>3</sup><br>*                   | TWA: 100 ppm<br>TWA: 130 mg/m <sup>3</sup><br>H*                                       | TWA: 100 ppm<br>TWA: 130 mg/m <sup>3</sup><br>Peak: 200 ppm<br>Peak: 260 mg/m <sup>3</sup><br>*  | TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup><br>STEL: 250 ppm<br>STEL: 325 mg/m <sup>3</sup><br>skin - potential for cutaneous absorption | TWA: 260 mg/m <sup>3</sup><br>*                                                                                                                                                                                                       |
| Acetic acid<br>64-19-7 | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 20 ppm<br>STEL: 50 mg/m <sup>3</sup>                              | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup>                                               | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>Peak: 20 ppm<br>Peak: 50 mg/m <sup>3</sup>           | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 15 ppm<br>STEL: 37 mg/m <sup>3</sup>                                                  | TWA: 25 mg/m <sup>3</sup><br>STEL: 50 mg/m <sup>3</sup>                                                                                                                                                                               |
| Chemical name          | Ireland                                                                                                             | Italy                                                                                  | Italy REL                                                                                        | Latvia                                                                                                                                  | Lithuania                                                                                                                                                                                                                             |
| Methanol<br>67-56-1    | TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup><br>STEL: 600 ppm<br>STEL: 780 mg/m <sup>3</sup><br>Sk*                   | TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup><br>pelle*                                   | TWA: 200 ppm<br>TWA: 262 mg/m <sup>3</sup><br>STEL: 250 ppm<br>STEL: 328 mg/m <sup>3</sup><br>*  | TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup><br>*                                                                                         | *<br>TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup>                                                                                                                                                                                       |
| Acetic acid<br>64-19-7 | TWA: 20 ppm<br>TWA: 50 mg/m <sup>3</sup><br>STEL: 20 ppm<br>STEL: 50 mg/m <sup>3</sup>                              | TWA: 25 ppm<br>TWA: 10 mg/m <sup>3</sup><br>STEL: 50 mg/m <sup>3</sup><br>STEL: 20 ppm | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 15 ppm<br>STEL: 37 mg/m <sup>3</sup>           | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 50 mg/m <sup>3</sup><br>STEL: 20 ppm                                                  | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 50 mg/m <sup>3</sup><br>STEL: 20 ppm                                                                                                                                                |
| Chemical name          | Luxembourg                                                                                                          | Malta                                                                                  | Netherlands                                                                                      | Norway                                                                                                                                  | Poland                                                                                                                                                                                                                                |
| Methanol<br>67-56-1    | *<br>TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup>                                                                     | *<br>TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup>                                        | TWA: 133 mg/m <sup>3</sup><br>H*                                                                 | TWA: 100 ppm<br>TWA: 130 mg/m <sup>3</sup><br>STEL: 150 ppm<br>STEL: 162.5 mg/m <sup>3</sup><br>H*                                      | STEL: 300 mg/m <sup>3</sup><br>TWA: 100 mg/m <sup>3</sup><br>Prohibited - substances or mixtures containing Methanol in weight concentration >3%; except fuels used in the model building, powerboating, fuel cells and biofuels<br>* |
| Acetic acid<br>64-19-7 | STEL: 50 mg/m <sup>3</sup><br>STEL: 20 ppm<br>TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup>                              | STEL: 20 ppm<br>STEL: 50 mg/m <sup>3</sup><br>TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup> | TWA: 25 mg/m <sup>3</sup><br>STEL: 50 mg/m <sup>3</sup>                                          | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 20 ppm<br>STEL: 50 mg/m <sup>3</sup>                                                  | STEL: 50 mg/m <sup>3</sup><br>TWA: 25 mg/m <sup>3</sup>                                                                                                                                                                               |
| Chemical name          | Portugal                                                                                                            | Romania                                                                                | Slovakia                                                                                         | Slovenia                                                                                                                                | Spain                                                                                                                                                                                                                                 |
| Methanol<br>67-56-1    | TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup><br>STEL: 250 ppm<br>P*                                                   | TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup><br>*                                        | TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup><br>*                                                  | TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup><br>STEL: 800 ppm<br>STEL: 1040 mg/m <sup>3</sup><br>*                                        | TWA: 200 ppm<br>TWA: 266 mg/m <sup>3</sup><br>vía dérmica*                                                                                                                                                                            |
| Acetic acid<br>64-19-7 | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 20 ppm<br>STEL: 50 mg/m <sup>3</sup>                              | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 20 ppm<br>STEL: 50 mg/m <sup>3</sup> | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>Ceiling: 50 mg/m <sup>3</sup>                        | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 50 mg/m <sup>3</sup><br>STEL: 20 ppm                                                  | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 20 ppm<br>STEL: 50 mg/m <sup>3</sup>                                                                                                                                                |
| Chemical name          | Sweden                                                                                                              |                                                                                        | Switzerland                                                                                      |                                                                                                                                         | United Kingdom                                                                                                                                                                                                                        |
| Methanol<br>67-56-1    | NGV: 200 ppm<br>NGV: 250 mg/m <sup>3</sup><br>Vägledande KGV: 250 ppm<br>Vägledande KGV: 350 mg/m <sup>3</sup><br>* |                                                                                        | TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup><br>STEL: 400 ppm<br>STEL: 520 mg/m <sup>3</sup><br>H* |                                                                                                                                         | TWA: 200 ppm<br>TWA: 266 mg/m <sup>3</sup><br>STEL: 250 ppm<br>STEL: 333 mg/m <sup>3</sup><br>Sk*                                                                                                                                     |
| Acetic acid<br>64-19-7 | NGV: 5 ppm<br>NGV: 13 mg/m <sup>3</sup><br>Bindande KGV: 10 ppm<br>Bindande KGV: 25 mg/m <sup>3</sup>               |                                                                                        | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 20 ppm<br>STEL: 50 mg/m <sup>3</sup>           |                                                                                                                                         | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 20 ppm<br>STEL: 50 mg/m <sup>3</sup>                                                                                                                                                |

| Chemical name       | European Union                                                                                                                                     | Austria                                   | Bulgaria                                                                                                                                                                                      | Croatia                                                                                                                                         | Czech Republic                                                                                                                              |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Methanol<br>67-56-1 | -                                                                                                                                                  | -                                         | -                                                                                                                                                                                             | 7.0 mg/g Creatinine - urine (Methanol) - at the end of the work shift                                                                           | 0.47 mmol/L (urine - Methanol end of shift)<br>15 mg/L (urine - Methanol end of shift)                                                      |
| Chemical name       | Denmark                                                                                                                                            | Finland                                   | France                                                                                                                                                                                        | Germany                                                                                                                                         | Germany                                                                                                                                     |
| Methanol<br>67-56-1 | -                                                                                                                                                  | -                                         | 15 mg/L - urine (Methanol) - end of shift                                                                                                                                                     | 15 mg/L - urine (Methanol) - end of shift<br>15 mg/L - urine (Methanol) - for long-term exposures: at the end of the shift after several shifts | 15 mg/L (urine - Methanol end of shift)<br>15 mg/L (urine - Methanol for long-term exposures: at the end of the shift after several shifts) |
| Chemical name       | Hungary                                                                                                                                            | Ireland                                   | Italy                                                                                                                                                                                         | Italy REL                                                                                                                                       |                                                                                                                                             |
| Methanol<br>67-56-1 | 30 mg/L (urine - Methanol end of shift)<br>940 µmol/L (urine - Methanol end of shift)                                                              | 15 mg/L - urine (Methanol) - end of shift | -                                                                                                                                                                                             | 15 mg/L - urine (Methanol) - end of shift                                                                                                       |                                                                                                                                             |
| Chemical name       | Latvia                                                                                                                                             | Luxembourg                                | Romania                                                                                                                                                                                       | Slovakia                                                                                                                                        |                                                                                                                                             |
| Methanol<br>67-56-1 | -                                                                                                                                                  | -                                         | 6 mg/L - urine (Methanol) - end of shift                                                                                                                                                      | 30 mg/L (urine - Methanol end of exposure or work shift)<br>30 mg/L (urine - Methanol after all work shifts)                                    |                                                                                                                                             |
| Chemical name       | Slovenia                                                                                                                                           | Spain                                     | Switzerland                                                                                                                                                                                   | United Kingdom                                                                                                                                  |                                                                                                                                             |
| Methanol<br>67-56-1 | 15 mg/L - urine (Methanol) - at the end of the work shift; for long-term exposure: at the end of the work shift after several consecutive workdays | 15 mg/L (urine - Methanol end of shift)   | 30 mg/L (urine - Methanol end of shift, and after several shifts (for long-term exposures))<br>936 µmol/L (urine - Methanol end of shift, and after several shifts (for long-term exposures)) | -                                                                                                                                               |                                                                                                                                             |

**Derived No Effect Level (DNEL)** No information available.

**Predicted No Effect Concentration (PNEC)**

## 8.2. Exposure controls

### Personal protective equipment

**Eye/face protection** Tight sealing safety goggles. Face protection shield.

**Hand protection** Wear suitable gloves. Impervious gloves.

**Skin and body protection** Wear suitable protective clothing. Long sleeved clothing. Chemical resistant apron. Antistatic boots.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**General hygiene considerations** Do not eat, drink or smoke when using this product. Contaminated work clothing should not be allowed out of the workplace. Regular cleaning of equipment, work area and clothing is recommended. Wash hands before breaks and immediately after handling the product. Wear suitable gloves and eye/face protection. Avoid contact with skin, eyes or clothing.

Remove and wash contaminated clothing and gloves, including the inside, before re-use.

Environmental exposure controls No information available.

## SECTION 9: Physical and chemical properties

### 9.1. Information on basic physical and chemical properties

|                 |                          |
|-----------------|--------------------------|
| Physical state  | Liquid                   |
| Appearance      | aqueous solution         |
| Colour          | blue                     |
| Odour           | Odourless.               |
| Odour threshold | No information available |

| <u>Property</u>                        | <u>Values</u>            | <u>Remarks • Method</u>  |
|----------------------------------------|--------------------------|--------------------------|
| Melting point / freezing point         | No data available        | None known               |
| Boiling point / boiling range          | 64.7 °C                  |                          |
| Flammability (solid, gas)              | No data available        | None known               |
| Flammability Limit in Air              |                          | None known               |
| Upper flammability or explosive limits | No data available        |                          |
| Lower flammability or explosive limits | No data available        |                          |
| Flash point                            | 35 °C                    |                          |
| Autoignition temperature               | No data available        |                          |
| Decomposition temperature              |                          | None known               |
| pH                                     | 1                        |                          |
| pH (as aqueous solution)               | No data available        | No information available |
| Kinematic viscosity                    | No data available        | None known               |
| Dynamic viscosity                      | No data available        | None known               |
| Water solubility                       | Miscible in water        |                          |
| Solubility(ies)                        | No data available        | None known               |
| Partition coefficient                  | No data available        | None known               |
| Vapour pressure                        | No data available        | None known               |
| Relative density                       | No data available        | None known               |
| Bulk density                           | No data available        |                          |
| Liquid Density                         | No data available        |                          |
| Vapour density                         | No data available        | None known               |
| Particle characteristics               |                          |                          |
| Particle Size                          | No information available |                          |
| Particle Size Distribution             | No information available |                          |

### 9.2. Other information

#### 9.2.1. Information with regard to physical hazard classes

Not applicable

#### 9.2.2. Other safety characteristics

No information available

## SECTION 10: Stability and reactivity

### 10.1. Reactivity

Reactivity No information available.

### 10.2. Chemical stability

Stability Stable under normal conditions.

#### Explosion data

Sensitivity to mechanical impact None.

Sensitivity to static discharge Yes.

### 10.3. Possibility of hazardous reactions

**Possibility of hazardous reactions** None under normal processing.

#### 10.4. Conditions to avoid

**Conditions to avoid** Heat, flames and sparks. Exposure to air or moisture over prolonged periods. Excessive heat.

#### 10.5. Incompatible materials

**Incompatible materials** Oxidizing agent. Acids. Bases.

#### 10.6. Hazardous decomposition products

**Hazardous decomposition products** None known based on information supplied.

### SECTION 11: Toxicological information

#### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

##### Information on likely routes of exposure

##### Product Information

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>   | Specific test data for the substance or mixture is not available. Corrosive by inhalation. (based on components). Inhalation of corrosive fumes/gases may cause coughing, choking, headache, dizziness, and weakness for several hours. Pulmonary edema may occur with tightness in the chest, shortness of breath, bluish skin, decreased blood pressure, and increased heart rate. Inhaled corrosive substances can lead to a toxic edema of the lungs. Pulmonary edema can be fatal. Harmful by inhalation.            |
| <b>Eye contact</b>  | Specific test data for the substance or mixture is not available. Causes serious eye damage. (based on components). Corrosive to the eyes and may cause severe damage including blindness. May cause irreversible damage to eyes.                                                                                                                                                                                                                                                                                         |
| <b>Skin contact</b> | Specific test data for the substance or mixture is not available. Corrosive. (based on components). Causes burns. Toxic in contact with skin.                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Ingestion</b>    | Specific test data for the substance or mixture is not available. Causes burns. (based on components). Ingestion causes burns of the upper digestive and respiratory tracts. May cause severe burning pain in the mouth and stomach with vomiting and diarrhea of dark blood. Blood pressure may decrease. Brownish or yellowish stains may be seen around the mouth. Swelling of the throat may cause shortness of breath and choking. May cause lung damage if swallowed. May be fatal if swallowed and enters airways. |

##### Symptoms related to the physical, chemical and toxicological characteristics

**Symptoms** Redness. Burning. May cause blindness. Coughing and/ or wheezing.

##### Acute toxicity

##### Numerical measures of toxicity

The following values are calculated based on chapter 3.1 of the GHS document

|                               |              |
|-------------------------------|--------------|
| ATEmix (oral)                 | 288.60 mg/kg |
| ATEmix (dermal)               | 799.40 mg/kg |
| ATEmix (inhalation-dust/mist) | 1.44 mg/l    |
| ATEmix (inhalation-vapor)     | 107.70 mg/l  |

##### Component Information

| Chemical name | Oral LD50            | Dermal LD50              | Inhalation LC50         |
|---------------|----------------------|--------------------------|-------------------------|
| Methanol      | = 6200 mg/kg ( Rat ) | = 15840 mg/kg ( Rabbit ) | = 22500 ppm ( Rat ) 8 h |

|             |                      |                         |                         |
|-------------|----------------------|-------------------------|-------------------------|
|             |                      |                         |                         |
| Acetic acid | = 3310 mg/kg ( Rat ) | = 1060 mg/kg ( Rabbit ) | = 11.4 mg/L ( Rat ) 4 h |

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                                          |                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin corrosion/irritation</b>         | Classification based on data available for ingredients.                                                                                                                                                                                                                                                                                             |
| <b>Serious eye damage/eye irritation</b> | Classification based on data available for ingredients. Causes burns.                                                                                                                                                                                                                                                                               |
| <b>Respiratory or skin sensitization</b> | No information available.                                                                                                                                                                                                                                                                                                                           |
| <b>Germ cell mutagenicity</b>            | No information available.                                                                                                                                                                                                                                                                                                                           |
| <b>Carcinogenicity</b>                   | No information available.                                                                                                                                                                                                                                                                                                                           |
| <b>Reproductive toxicity</b>             | No information available.                                                                                                                                                                                                                                                                                                                           |
| <b>STOT - single exposure</b>            | Based on the classification criteria of the Globally Harmonized System as adopted in the country or region with which this safety data sheet complies, this product has been determined to cause systemic target organ toxicity from acute exposure. (STOT SE). Causes damage to organs if swallowed. Causes damage to organs in contact with skin. |
| <b>STOT - repeated exposure</b>          | No information available.                                                                                                                                                                                                                                                                                                                           |
| <b>Aspiration hazard</b>                 | No information available.                                                                                                                                                                                                                                                                                                                           |

**11.2. Information on other hazards****11.2.1. Endocrine disrupting properties**

|                                        |                           |
|----------------------------------------|---------------------------|
| <b>Endocrine disrupting properties</b> | No information available. |
|----------------------------------------|---------------------------|

**11.2.2. Other information**

|                              |                           |
|------------------------------|---------------------------|
| <b>Other adverse effects</b> | No information available. |
|------------------------------|---------------------------|

**SECTION 12: Ecological information****12.1. Toxicity**

|                                 |                                                                             |
|---------------------------------|-----------------------------------------------------------------------------|
| <b>Ecotoxicity</b>              | Harmful to aquatic life.                                                    |
| <b>Unknown aquatic toxicity</b> | Contains 0 % of components with unknown hazards to the aquatic environment. |

| Chemical name | Algae/aquatic plants | Fish                                                                                                                                                                      | Toxicity to microorganisms | Crustacea |
|---------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-----------|
| Methanol      | -                    | LC50: 13500 - 17600mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: 18 - 20mL/L (96h, <i>Oncorhynchus mykiss</i> )<br>LC50: 19500 - 20700mg/L (96h, <i>Oncorhynchus</i> ) | -                          | -         |

|             |   |                                                                                                           |   |                                       |
|-------------|---|-----------------------------------------------------------------------------------------------------------|---|---------------------------------------|
|             |   | mykiss)<br>LC50: =28200mg/L (96h,<br>Pimephales promelas)<br>LC50: >100mg/L (96h,<br>Pimephales promelas) |   |                                       |
| Acetic acid | - | LC50: =75mg/L (96h,<br>Lepomis macrochirus)<br>LC50: =79mg/L (96h,<br>Pimephales promelas)                | - | EC50: =65mg/L (48h,<br>Daphnia magna) |

**12.2. Persistence and degradability**

**Persistence and degradability** No information available.

**12.3. Bioaccumulative potential****Bioaccumulation****Component Information**

| Chemical name | Partition coefficient |
|---------------|-----------------------|
| Methanol      | -0.77                 |
| Acetic acid   | -0.17                 |

**12.4. Mobility in soil**

**Mobility in soil** No information available.

**12.5. Results of PBT and vPvB assessment****PBT and vPvB assessment**

| Chemical name | PBT and vPvB assessment                                                                                                        |
|---------------|--------------------------------------------------------------------------------------------------------------------------------|
| Methanol      | The substance is not PBT / vPvB PBT assessment does not apply Further information relevant for the PBT assessment is necessary |
| Acetic acid   | The substance is not PBT / vPvB PBT assessment does not apply                                                                  |

**12.6. Endocrine disrupting properties**

**Endocrine disrupting properties** No information available.

**12.7. Other adverse effects**

No information available.

## SECTION 13: Disposal considerations

**13.1. Waste treatment methods****Waste from residues/unused products**

Should not be released into the environment. Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging**

Empty containers pose a potential fire and explosion hazard. Do not cut, puncture or weld containers.

## SECTION 14: Transport information

**IATA**

|                                    |                                                                                         |
|------------------------------------|-----------------------------------------------------------------------------------------|
| 14.1 UN number or ID number        | UN2924                                                                                  |
| 14.2 UN proper shipping name       | Flammable liquid, corrosive, n.o.s (Methanol Solution, Acetic acid)                     |
| 14.3 Transport hazard class(es)    | 3                                                                                       |
| Subsidiary hazard class            | 8                                                                                       |
| 14.4 Packing group                 | III                                                                                     |
| Description                        | UN2924, Flammable liquid, corrosive, n.o.s (Methanol Solution, Acetic acid), 3 (8), III |
| 14.5 Environmental hazards         | Not applicable                                                                          |
| 14.6 Special Precautions for Users |                                                                                         |
| Special Provisions                 | A3, A803                                                                                |

**IMDG**

|                                                              |                                                                                                       |
|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 14.1 UN number or ID number                                  | UN2924                                                                                                |
| 14.2 UN proper shipping name                                 | FLAMMABLE LIQUID, CORROSIVE, N.O.S. (Methanol Solution, Acetic acid)                                  |
| 14.3 Transport hazard class(es)                              | 3                                                                                                     |
| Subsidiary hazard class                                      | 8                                                                                                     |
| 14.4 Packing group                                           | III                                                                                                   |
| Description                                                  | UN2924, FLAMMABLE LIQUID, CORROSIVE, N.O.S. (Methanol Solution, Acetic acid), 3 (8), III, (35°C C.C.) |
| 14.5 Environmental hazards                                   | Not applicable                                                                                        |
| 14.6 Special Precautions for Users                           |                                                                                                       |
| Special Provisions                                           | 223, 274                                                                                              |
| EmS-No                                                       | F-E, S-C                                                                                              |
| 14.7 Maritime transport in bulk according to IMO instruments | No information available                                                                              |

**RID**

|                                    |                                                                                          |
|------------------------------------|------------------------------------------------------------------------------------------|
| 14.1 UN number                     | UN2924                                                                                   |
| 14.2 UN proper shipping name       | FLAMMABLE LIQUID, CORROSIVE, N.O.S. (Methanol Solution, Acetic acid)                     |
| 14.3 Transport hazard class(es)    | 3                                                                                        |
| Subsidiary hazard class            | 8                                                                                        |
| 14.4 Packing group                 | III                                                                                      |
| Description                        | UN2924, FLAMMABLE LIQUID, CORROSIVE, N.O.S. (Methanol Solution, Acetic acid), 3 (8), III |
| 14.5 Environmental hazards         | Not applicable                                                                           |
| 14.6 Special Precautions for Users |                                                                                          |
| Special Provisions                 | 274                                                                                      |
| Classification code                | FC                                                                                       |

**ADR**

|                                    |                                                                                        |
|------------------------------------|----------------------------------------------------------------------------------------|
| 14.1 UN number or ID number        | 2924                                                                                   |
| 14.2 UN proper shipping name       | FLAMMABLE LIQUID, CORROSIVE, N.O.S. (Methanol Solution, Acetic acid)                   |
| 14.3 Transport hazard class(es)    | 3                                                                                      |
| Subsidiary class                   | 8                                                                                      |
| 14.4 Packing group                 | III                                                                                    |
| Description                        | 2924, FLAMMABLE LIQUID, CORROSIVE, N.O.S. (Methanol Solution, Acetic acid), 3 (8), III |
| 14.5 Environmental hazards         | Not applicable                                                                         |
| 14.6 Special Precautions for Users |                                                                                        |
| Special Provisions                 | 274                                                                                    |
| Classification code                | FC                                                                                     |
| Tunnel restriction code            | (D/E)                                                                                  |

**SECTION 15: Regulatory information****15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture****National regulations****France****Occupational Illnesses (R-463-3, France)**

| Chemical name       | French RG number | Title |
|---------------------|------------------|-------|
| Methanol<br>67-56-1 | RG 84            | -     |

**Germany****Water hazard class (WGK)**

obviously hazardous to water (WGK 2)

**European Union**

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work.

**Authorizations and/or restrictions on use:**

This product contains one or more substance(s) subject to restriction (Regulation (EC) No. 1907/2006 (REACH), Annex XVII)

| Chemical name         | Restricted substance per REACH Annex XVII | Substance subject to authorization per REACH Annex XIV |
|-----------------------|-------------------------------------------|--------------------------------------------------------|
| Methanol - 67-56-1    | 69.<br>75.                                | -                                                      |
| Acetic acid - 64-19-7 | 75.                                       | -                                                      |

**Persistent Organic Pollutants**

Not applicable

**Dangerous substance category per Seveso Directive (2012/18/EU)**

H3 - STOT SPECIFIC TARGET ORGAN TOXICITY - SINGLE EXPOSURE

P5a - FLAMMABLE LIQUIDS

P5b - FLAMMABLE LIQUIDS

P5c - FLAMMABLE LIQUIDS

**Named dangerous substances per Seveso Directive (2012/18/EU)**

| Chemical name      | Lower-tier requirements (tons) | Upper-tier requirements (tons) |
|--------------------|--------------------------------|--------------------------------|
| Methanol - 67-56-1 | 500                            | 5000                           |

**Ozone-depleting substances (ODS) regulation (EC) 1005/2009**

Not applicable

**Plant protection products directive (91/414/EEC)**

| Chemical name         | Plant protection products directive (91/414/EEC) |
|-----------------------|--------------------------------------------------|
| Acetic acid - 64-19-7 | Plant protection agent                           |

**EU - Biocides****International Inventories**

Contact supplier for inventory compliance status

**15.2. Chemical safety assessment****Chemical Safety Report**

No information available

**SECTION 16: Other information****Key or legend to abbreviations and acronyms used in the safety data sheet****Full text of H-Statements referred to under section 3**

H225 - Highly flammable liquid and vapor

H226 - Flammable liquid and vapor

H301 - Toxic if swallowed

H311 - Toxic in contact with skin

H314 - Causes severe skin burns and eye damage

H331 - Toxic if inhaled

H370 - Causes damage to organs

**Legend**

SVHC: Substances of Very High Concern for Authorisation:

**Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |

| Classification procedure                                        |                       |
|-----------------------------------------------------------------|-----------------------|
| Classification according to Regulation (EC) No. 1272/2008 [CLP] | Method Used           |
| Acute oral toxicity                                             | Calculation method    |
| Acute dermal toxicity                                           | Calculation method    |
| Acute inhalation toxicity - gas                                 | Calculation method    |
| Acute inhalation toxicity - Vapour                              | Calculation method    |
| Acute inhalation toxicity - dust/mist                           | Calculation method    |
| Skin corrosion/irritation                                       | Calculation method    |
| Serious eye damage/eye irritation                               | Calculation method    |
| Respiratory sensitisation                                       | Calculation method    |
| Skin sensitisation                                              | Calculation method    |
| Mutagenicity                                                    | Calculation method    |
| Carcinogenicity                                                 | Calculation method    |
| Reproductive toxicity                                           | Calculation method    |
| STOT - single exposure                                          | Calculation method    |
| STOT - repeated exposure                                        | Calculation method    |
| Acute aquatic toxicity                                          | Calculation method    |
| Chronic aquatic toxicity                                        | Calculation method    |
| Aspiration hazard                                               | Calculation method    |
| Ozone                                                           | Calculation method    |
| Corrosive to metals                                             | On basis of test data |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 European Chemicals Agency (ECHA) Committee for Risk Assessment (ECHA\_RAC)  
 European Chemicals Agency (ECHA) (ECHA\_API)  
 EPA (Environmental Protection Agency)  
 Acute Exposure Guideline Level(s) (AEGL(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
 Organization for Economic Co-operation and Development Screening Information Data Set  
 World Health Organization

**Revision Note** Reformatted and updated existing information

**Revision date** 31-Aug-2022

This material safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
Regulation (EC) No. 1907/2006 and Regulation (EC) No. 1272/2008

Revision date 31-Aug-2022

Revision Number 1.3

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### 1.1. Product identifier

**Product Name** Coomassie Brilliant Blue R-250 Destain Solution

**Catalogue Number(s)** 1610438, 1610439, 1610438EDU, 1610439EDU

**Pure substance/mixture** Mixture

Contains Methanol

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

**Recommended use** Laboratory chemicals

**Uses advised against** No information available

### 1.3. Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories Ltd  
The Junction  
Station Road  
Watford, WD17 1ET  
UK

Bio-Rad Laboratories Pvt. Ltd.  
Bio-Rad House  
86-87, Udyog Vihar Phase IV Gurgaon  
122005  
Haryana India

Bio-Rad Laboratories (Pty) Ltd.  
34 Bolton Road  
Parkwood, Johannesburg 2193  
South Africa

For further information, please contact

#### **Technical Service**

00800 00246 723  
Ireland: Techsupport.UK@bio-rad.com  
India: support.india@bio-rad.com  
South Africa: cdg\_techsupport\_eemea@bio-rad.com

### 1.4. Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC Ireland: 353-19014670  
CHEMTREC India: 000-800-100-7141  
CHEMTREC South Africa: 0-800-983-611

## SECTION 2: Hazards identification

### 2.1. Classification of the substance or mixture

Regulation (EC) No 1272/2008

**Acute toxicity - Oral**

Category 3 - (H301)

|                                                  |                     |
|--------------------------------------------------|---------------------|
| Acute toxicity - Dermal                          | Category 3 - (H311) |
| Acute toxicity - Inhalation (Dusts/Mists)        | Category 4 - (H332) |
| Skin corrosion/irritation                        | Category 1 - (H314) |
| Serious eye damage/eye irritation                | Category 1 - (H318) |
| Specific target organ toxicity (single exposure) | Category 1          |
| Corrosive to metals                              | Category 1          |
| Flammable liquids                                | Category 3 - (H226) |

## 2.2. Label elements

Contains Methanol



Signal word

Danger

## Hazard statements

H301 - Toxic if swallowed

H311 - Toxic in contact with skin

H314 - Causes severe skin burns and eye damage

H332 - Harmful if inhaled

H370 - Causes damage to organs

H290 - May be corrosive to metals

H226 - Flammable liquid and vapor

## Precautionary Statements - EU (§28, 1272/2008)

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water [or shower]

P331 - Do NOT induce vomiting

P370 + P378 - In case of fire: Use dry chemical, CO<sub>2</sub>, water spray or alcohol-resistant foam to extinguish

P403 + P235 - Store in a well-ventilated place. Keep cool

P501 - Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

P301 + P330 + P331 - IF SWALLOWED: Rinse mouth. Do NOT induce vomiting

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

## 2.3. Other hazards

Harmful to aquatic life.

## SECTION 3: Composition/information on ingredients

### 3.1 Substances

Not applicable

### 3.2 Mixtures

| Chemical name       | Weight-% | REACH registration number | EC No     | Classification according to Regulation (EC) No. 1272/2008 [CLP]                       | Specific concentration limit (SCL) | M-Factor | M-Factor (long-term) |
|---------------------|----------|---------------------------|-----------|---------------------------------------------------------------------------------------|------------------------------------|----------|----------------------|
| Methanol<br>67-56-1 | 20 - 35  | No data available         | 200-659-6 | Acute Tox. 3 (H301)<br>Acute Tox. 3 (H311)<br>Acute Tox. 3 (H331)<br>STOT SE 1 (H370) | STOT SE 1 ::<br>C>=1%              | -        | -                    |

|                        |         |                   |           |                                             |                                                                                                                             |   |   |
|------------------------|---------|-------------------|-----------|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---|---|
|                        |         |                   |           | Flam. Liq. 2 (H225)                         |                                                                                                                             |   |   |
| Acetic acid<br>64-19-7 | 10 - 20 | No data available | 200-580-7 | Skin Corr. 1A (H314)<br>Flam. Liq. 3 (H226) | Eye Irrit. 2 ::<br>10%≤C<25%<br>Skin Corr. 1A ::<br>C≥90%<br>Skin Corr. 1B ::<br>25%≤C<90%<br>Skin Irrit. 2 ::<br>10%≤C<25% | - | - |

**Full text of H- and EUH-phrases: see section 16**

#### Acute Toxicity Estimate

If LD50/LC50 data is not available or does not correspond to the classification category, then the appropriate conversion value from CLP Annex I, Table 3.1.2, is used to calculate the acute toxicity estimate (ATEmix) for classifying a mixture based on its components

| Chemical name          | Oral LD50 mg/kg | Dermal LD50 mg/kg | Inhalation LC50 - 4 hour - dust/mist - mg/L | Inhalation LC50 - 4 hour - vapor - mg/L | Inhalation LC50 - 4 hour - gas - ppm |
|------------------------|-----------------|-------------------|---------------------------------------------|-----------------------------------------|--------------------------------------|
| Methanol<br>67-56-1    | 6200            | 15840             | No data available                           | 41.6976                                 | No data available                    |
| Acetic acid<br>64-19-7 | 3310            | 1060              | 11.4                                        | No data available                       | No data available                    |

This product does not contain candidate substances of very high concern at a concentration ≥0.1% (Regulation (EC) No. 1907/2006 (REACH), Article 59)

## SECTION 4: First aid measures

### 4.1. Description of first aid measures

|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b>                     | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Inhalation</b>                         | Remove to fresh air. If breathing has stopped, give artificial respiration. Get medical attention immediately. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. If breathing is difficult, (trained personnel should) give oxygen. Delayed pulmonary edema may occur. Get immediate medical advice/attention. |
| <b>Eye contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Keep eye wide open while rinsing. Do not rub affected area. Remove contact lenses, if present and easy to do. Continue rinsing. Get immediate medical advice/attention.                                                                                                                                                                                                               |
| <b>Skin contact</b>                       | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Get immediate medical advice/attention.                                                                                                                                                                                                                                                                                                                                  |
| <b>Ingestion</b>                          | Rinse mouth. Never give anything by mouth to an unconscious person. Do NOT induce vomiting. Get immediate medical advice/attention.                                                                                                                                                                                                                                                                                                                                            |
| <b>Self-protection of the first aider</b> | Remove all sources of ignition. Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. Use personal protective equipment as required. See section 8 for more information. Avoid contact with skin, eyes or clothing. Avoid direct contact with skin. Use barrier to give mouth-to-mouth resuscitation. Avoid breathing vapors or mists.                                              |

### 4.2. Most important symptoms and effects, both acute and delayed

|                 |                                                                        |
|-----------------|------------------------------------------------------------------------|
| <b>Symptoms</b> | Burning sensation. Coughing and/ or wheezing. Difficulty in breathing. |
|-----------------|------------------------------------------------------------------------|

### 4.3. Indication of any immediate medical attention and special treatment needed

|                           |                                                                                                                                                                                                                                                                                                                                        |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Note to physicians</b> | Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Do not give chemical antidotes. Asphyxia from glottal edema may occur. Marked decrease in blood pressure may occur with moist rales, frothy sputum, and high pulse pressure. |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## SECTION 5: Firefighting measures

### 5.1. Extinguishing media

**Suitable Extinguishing Media** Dry chemical. Carbon dioxide (CO<sub>2</sub>). Water spray. Alcohol resistant foam.

**Large Fire** CAUTION: Use of water spray when fighting fire may be inefficient.

**Unsuitable extinguishing media** Do not scatter spilled material with high pressure water streams.

### 5.2. Special hazards arising from the substance or mixture

**Specific hazards arising from the chemical** Risk of ignition. Keep product and empty container away from heat and sources of ignition. In the event of fire, cool tanks with water spray. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. The product causes burns of eyes, skin and mucous membranes. Thermal decomposition can lead to release of irritating gases and vapors.

### 5.3. Advice for firefighters

**Special protective equipment and precautions for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

## SECTION 6: Accidental release measures

### 6.1. Personal precautions, protective equipment and emergency procedures

**Personal precautions** Evacuate personnel to safe areas. Use personal protective equipment as required. See section 8 for more information. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Keep people away from and upwind of spill/leak. ELIMINATE all ignition sources (no smoking, flares, sparks or flames in immediate area). Pay attention to flashback. Take precautionary measures against static discharges. All equipment used when handling the product must be grounded. Do not touch or walk through spilled material. Attention! Corrosive material. Avoid breathing vapors or mists.

**Other information** Ventilate the area. Refer to protective measures listed in Sections 7 and 8.

**For emergency responders** Use personal protection recommended in Section 8.

### 6.2. Environmental precautions

**Environmental precautions** Refer to protective measures listed in Sections 7 and 8. Prevent further leakage or spillage if safe to do so. Prevent product from entering drains. Should not be released into the environment. Do not allow to enter into soil/subsoil.

### 6.3. Methods and material for containment and cleaning up

**Methods for containment** Stop leak if you can do it without risk. Do not touch or walk through spilled material. A vapor suppressing foam may be used to reduce vapors. Dike far ahead of spill to collect runoff water. Keep out of drains, sewers, ditches and waterways. Absorb with earth, sand or other non-combustible material and transfer to containers for later disposal.

**Methods for cleaning up** Take precautionary measures against static discharges. Dam up. Soak up with inert absorbent material. Pick up and transfer to properly labeled containers.

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

### 6.4. Reference to other sections

**Reference to other sections** See section 8 for more information. See section 13 for more information.

## SECTION 7: Handling and storage

### 7.1. Precautions for safe handling

#### Advice on safe handling

Use personal protection equipment. Avoid breathing vapors or mists. Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. Use grounding and bonding connection when transferring this material to prevent static discharge, fire or explosion. Use spark-proof tools and explosion-proof equipment. Keep in an area equipped with sprinklers. Use according to package label instructions. Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Take off contaminated clothing and wash before reuse. In case of insufficient ventilation, wear suitable respiratory equipment. Handle product only in closed system or provide appropriate exhaust ventilation. Do not eat, drink or smoke when using this product.

#### General hygiene considerations

Do not eat, drink or smoke when using this product. Contaminated work clothing should not be allowed out of the workplace. Regular cleaning of equipment, work area and clothing is recommended. Wash hands before breaks and immediately after handling the product. Wear suitable gloves and eye/face protection. Avoid contact with skin, eyes or clothing. Remove and wash contaminated clothing and gloves, including the inside, before re-use.

### 7.2. Conditions for safe storage, including any incompatibilities

#### Storage Conditions

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks, flame and other sources of ignition (i.e., pilot lights, electric motors and static electricity). Keep in properly labeled containers. Do not store near combustible materials. Keep in an area equipped with sprinklers. Store in accordance with the particular national regulations. Store in accordance with local regulations. Protect from moisture. Store locked up. Keep out of the reach of children. Store away from other materials. Store according to product and label instructions.

### 7.3. Specific end use(s)

**Risk Management Methods (RMM)** The information required is contained in this Safety Data Sheet.

## SECTION 8: Exposure controls/personal protection

### 8.1. Control parameters

#### Exposure Limits

| Chemical name          | European Union                                  | Austria                                                                                         | Belgium                                                                                         | Bulgaria                                                                                         | Croatia                                                                                            |
|------------------------|-------------------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Methanol<br>67-56-1    | TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup><br>* | TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup><br>STEL 800 ppm<br>STEL 1040 mg/m <sup>3</sup><br>H* | TWA: 200 ppm<br>TWA: 266 mg/m <sup>3</sup><br>STEL: 250 ppm<br>STEL: 333 mg/m <sup>3</sup><br>* | TWA: 200 ppm<br>TWA: 260.0 mg/m <sup>3</sup><br>K*                                               | TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup><br>*                                                    |
| Acetic acid<br>64-19-7 | -                                               | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL 20 ppm<br>STEL 50 mg/m <sup>3</sup>            | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 15 ppm<br>STEL: 38 mg/m <sup>3</sup>          | STEL: 50 mg/m <sup>3</sup><br>STEL: 20 ppm<br>TWA: 25 mg/m <sup>3</sup><br>TWA: 10 ppm           | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 20 ppm<br>STEL: 50 mg/m <sup>3</sup>             |
| Chemical name          | Cyprus                                          | Czech Republic                                                                                  | Denmark                                                                                         | Estonia                                                                                          | Finland                                                                                            |
| Methanol<br>67-56-1    | *<br>TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup> | TWA: 250 mg/m <sup>3</sup><br>Ceiling: 1000 mg/m <sup>3</sup><br>*                              | TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup><br>H*                                                | TWA: 200 ppm<br>TWA: 250 mg/m <sup>3</sup><br>STEL: 250 ppm<br>STEL: 350 mg/m <sup>3</sup><br>A* | TWA: 200 ppm<br>TWA: 270 mg/m <sup>3</sup><br>STEL: 250 ppm<br>STEL: 330 mg/m <sup>3</sup><br>iho* |
| Acetic acid            | STEL: 50 mg/m <sup>3</sup>                      | TWA: 25 mg/m <sup>3</sup>                                                                       | TWA: 10 ppm                                                                                     | TWA: 10 ppm                                                                                      | TWA: 5 ppm                                                                                         |

|                        |                                                                                                                     |                                                                                        |                                                                                                  |                                                                                                                                         |                                                                                                                                                                                                                                      |
|------------------------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 64-19-7                | STEL: 20 ppm<br>TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup>                                                            | Ceiling: 50 mg/m <sup>3</sup>                                                          | TWA: 25 mg/m <sup>3</sup>                                                                        | TWA: 25 mg/m <sup>3</sup><br>STEL: 10 ppm<br>STEL: 25 mg/m <sup>3</sup>                                                                 | TWA: 13 mg/m <sup>3</sup><br>STEL: 10 ppm<br>STEL: 25 mg/m <sup>3</sup>                                                                                                                                                              |
| Chemical name          | France                                                                                                              | Germany                                                                                | Germany MAK                                                                                      | Greece                                                                                                                                  | Hungary                                                                                                                                                                                                                              |
| Methanol<br>67-56-1    | TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup><br>STEL: 1000 ppm<br>STEL: 1300 mg/m <sup>3</sup><br>*                   | TWA: 100 ppm<br>TWA: 130 mg/m <sup>3</sup><br>H*                                       | TWA: 100 ppm<br>TWA: 130 mg/m <sup>3</sup><br>Peak: 200 ppm<br>Peak: 260 mg/m <sup>3</sup><br>*  | TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup><br>STEL: 250 ppm<br>STEL: 325 mg/m <sup>3</sup><br>skin - potential for cutaneous absorption | TWA: 260 mg/m <sup>3</sup><br>*                                                                                                                                                                                                      |
| Acetic acid<br>64-19-7 | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 20 ppm<br>STEL: 50 mg/m <sup>3</sup>                              | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup>                                               | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>Peak: 20 ppm<br>Peak: 50 mg/m <sup>3</sup>           | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 15 ppm<br>STEL: 37 mg/m <sup>3</sup>                                                  | TWA: 25 mg/m <sup>3</sup><br>STEL: 50 mg/m <sup>3</sup>                                                                                                                                                                              |
| Chemical name          | Ireland                                                                                                             | Italy                                                                                  | Italy REL                                                                                        | Latvia                                                                                                                                  | Lithuania                                                                                                                                                                                                                            |
| Methanol<br>67-56-1    | TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup><br>STEL: 600 ppm<br>STEL: 780 mg/m <sup>3</sup><br>Sk*                   | TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup><br>pelle*                                   | TWA: 200 ppm<br>TWA: 262 mg/m <sup>3</sup><br>STEL: 250 ppm<br>STEL: 328 mg/m <sup>3</sup><br>*  | TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup><br>*                                                                                         | *<br>TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup>                                                                                                                                                                                      |
| Acetic acid<br>64-19-7 | TWA: 20 ppm<br>TWA: 50 mg/m <sup>3</sup><br>STEL: 20 ppm<br>STEL: 50 mg/m <sup>3</sup>                              | TWA: 25 ppm<br>TWA: 10 mg/m <sup>3</sup><br>STEL: 50 mg/m <sup>3</sup><br>STEL: 20 ppm | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 15 ppm<br>STEL: 37 mg/m <sup>3</sup>           | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 50 mg/m <sup>3</sup><br>STEL: 20 ppm                                                  | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 50 mg/m <sup>3</sup><br>STEL: 20 ppm                                                                                                                                               |
| Chemical name          | Luxembourg                                                                                                          | Malta                                                                                  | Netherlands                                                                                      | Norway                                                                                                                                  | Poland                                                                                                                                                                                                                               |
| Methanol<br>67-56-1    | *<br>TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup>                                                                     | *<br>TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup>                                        | TWA: 133 mg/m <sup>3</sup><br>H*                                                                 | TWA: 100 ppm<br>TWA: 130 mg/m <sup>3</sup><br>STEL: 150 ppm<br>STEL: 162.5 mg/m <sup>3</sup><br>H*                                      | STEL: 300 mg/m <sup>3</sup><br>TWA: 100 mg/m <sup>3</sup><br>Prohibited - substances or mixtures containing Methanol in weight concentration >3%;except fuels used in the model building, powerboating, fuel cells and biofuels<br>* |
| Acetic acid<br>64-19-7 | STEL: 50 mg/m <sup>3</sup><br>STEL: 20 ppm<br>TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup>                              | STEL: 20 ppm<br>STEL: 50 mg/m <sup>3</sup><br>TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup> | TWA: 25 mg/m <sup>3</sup><br>STEL: 50 mg/m <sup>3</sup>                                          | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 20 ppm<br>STEL: 50 mg/m <sup>3</sup>                                                  | STEL: 50 mg/m <sup>3</sup><br>TWA: 25 mg/m <sup>3</sup>                                                                                                                                                                              |
| Chemical name          | Portugal                                                                                                            | Romania                                                                                | Slovakia                                                                                         | Slovenia                                                                                                                                | Spain                                                                                                                                                                                                                                |
| Methanol<br>67-56-1    | TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup><br>STEL: 250 ppm<br>P*                                                   | TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup><br>*                                        | TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup><br>*                                                  | TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup><br>STEL: 800 ppm<br>STEL: 1040 mg/m <sup>3</sup><br>*                                        | TWA: 200 ppm<br>TWA: 266 mg/m <sup>3</sup><br>vía dérmica*                                                                                                                                                                           |
| Acetic acid<br>64-19-7 | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 20 ppm<br>STEL: 50 mg/m <sup>3</sup>                              | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 20 ppm<br>STEL: 50 mg/m <sup>3</sup> | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>Ceiling: 50 mg/m <sup>3</sup>                        | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 50 mg/m <sup>3</sup><br>STEL: 20 ppm                                                  | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 20 ppm<br>STEL: 50 mg/m <sup>3</sup>                                                                                                                                               |
| Chemical name          | Sweden                                                                                                              |                                                                                        | Switzerland                                                                                      |                                                                                                                                         | United Kingdom                                                                                                                                                                                                                       |
| Methanol<br>67-56-1    | NGV: 200 ppm<br>NGV: 250 mg/m <sup>3</sup><br>Vägledande KGV: 250 ppm<br>Vägledande KGV: 350 mg/m <sup>3</sup><br>* |                                                                                        | TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup><br>STEL: 400 ppm<br>STEL: 520 mg/m <sup>3</sup><br>H* |                                                                                                                                         | TWA: 200 ppm<br>TWA: 266 mg/m <sup>3</sup><br>STEL: 250 ppm<br>STEL: 333 mg/m <sup>3</sup><br>Sk*                                                                                                                                    |
| Acetic acid<br>64-19-7 | NGV: 5 ppm<br>NGV: 13 mg/m <sup>3</sup><br>Bindande KGV: 10 ppm<br>Bindande KGV: 25 mg/m <sup>3</sup>               |                                                                                        | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 20 ppm<br>STEL: 50 mg/m <sup>3</sup>           |                                                                                                                                         | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 20 ppm<br>STEL: 50 mg/m <sup>3</sup>                                                                                                                                               |

**Biological occupational exposure limits**

| Chemical name       | European Union                                                                                                                                     | Austria    | Bulgaria                                  | Croatia                                                                                                                                                                                       | Czech Republic                                                                                                                              |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Methanol<br>67-56-1 | -                                                                                                                                                  | -          | -                                         | 7.0 mg/g Creatinine - urine (Methanol) - at the end of the work shift                                                                                                                         | 0.47 mmol/L (urine - Methanol end of shift)<br>15 mg/L (urine - Methanol end of shift)                                                      |
| Chemical name       | Denmark                                                                                                                                            | Finland    | France                                    | Germany                                                                                                                                                                                       | Germany                                                                                                                                     |
| Methanol<br>67-56-1 | -                                                                                                                                                  | -          | 15 mg/L - urine (Methanol) - end of shift | 15 mg/L - urine (Methanol) - end of shift<br>15 mg/L - urine (Methanol) - for long-term exposures: at the end of the shift after several shifts                                               | 15 mg/L (urine - Methanol end of shift)<br>15 mg/L (urine - Methanol for long-term exposures: at the end of the shift after several shifts) |
| Chemical name       | Hungary                                                                                                                                            |            | Ireland                                   | Italy                                                                                                                                                                                         | Italy REL                                                                                                                                   |
| Methanol<br>67-56-1 | 30 mg/L (urine - Methanol end of shift)<br>940 µmol/L (urine - Methanol end of shift)                                                              |            | 15 mg/L - urine (Methanol) - end of shift | -                                                                                                                                                                                             | 15 mg/L - urine (Methanol) - end of shift                                                                                                   |
| Chemical name       | Latvia                                                                                                                                             | Luxembourg |                                           | Romania                                                                                                                                                                                       | Slovakia                                                                                                                                    |
| Methanol<br>67-56-1 | -                                                                                                                                                  | -          |                                           | 6 mg/L - urine (Methanol) - end of shift                                                                                                                                                      | 30 mg/L (urine - Methanol end of exposure or work shift)<br>30 mg/L (urine - Methanol after all work shifts)                                |
| Chemical name       | Slovenia                                                                                                                                           |            | Spain                                     | Switzerland                                                                                                                                                                                   | United Kingdom                                                                                                                              |
| Methanol<br>67-56-1 | 15 mg/L - urine (Methanol) - at the end of the work shift; for long-term exposure: at the end of the work shift after several consecutive workdays |            | 15 mg/L (urine - Methanol end of shift)   | 30 mg/L (urine - Methanol end of shift, and after several shifts (for long-term exposures))<br>936 µmol/L (urine - Methanol end of shift, and after several shifts (for long-term exposures)) | -                                                                                                                                           |

**Derived No Effect Level (DNEL)** No information available.  
**Predicted No Effect Concentration (PNEC)**

**8.2. Exposure controls****Personal protective equipment**

**Eye/face protection** Tight sealing safety goggles. Face protection shield.

**Hand protection** Wear suitable gloves. Impervious gloves.

**Skin and body protection** Wear suitable protective clothing. Long sleeved clothing. Chemical resistant apron. Antistatic boots.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**General hygiene considerations** Do not eat, drink or smoke when using this product. Contaminated work clothing should not be allowed out of the workplace. Regular cleaning of equipment, work area and clothing is recommended. Wash hands before breaks and immediately after handling the product.

Wear suitable gloves and eye/face protection. Avoid contact with skin, eyes or clothing.  
Remove and wash contaminated clothing and gloves, including the inside, before re-use.

**Environmental exposure controls** No information available.

## SECTION 9: Physical and chemical properties

### 9.1. Information on basic physical and chemical properties

|                        |                          |
|------------------------|--------------------------|
| <b>Physical state</b>  | Liquid                   |
| <b>Appearance</b>      | Liquid aqueous solution  |
| <b>Colour</b>          | colourless               |
| <b>Odour</b>           | Acrid.                   |
| <b>Odour threshold</b> | No information available |

| <u>Property</u>                               | <u>Values</u>            | <u>Remarks • Method</u>  |
|-----------------------------------------------|--------------------------|--------------------------|
| <b>Melting point / freezing point</b>         | No data available        | None known               |
| <b>Boiling point / boiling range</b>          | No data available        | None known               |
| <b>Flammability (solid, gas)</b>              | No data available        | None known               |
| <b>Flammability Limit in Air</b>              |                          | None known               |
| <b>Upper flammability or explosive limits</b> | No data available        |                          |
| <b>Lower flammability or explosive limits</b> | No data available        |                          |
| <b>Flash point</b>                            | 35 °C                    |                          |
| <b>Autoignition temperature</b>               | 463 °C                   |                          |
| <b>Decomposition temperature</b>              |                          | None known               |
| <b>pH</b>                                     | <2                       |                          |
| <b>pH (as aqueous solution)</b>               | No data available        | No information available |
| <b>Kinematic viscosity</b>                    | No data available        | None known               |
| <b>Dynamic viscosity</b>                      | No data available        | None known               |
| <b>Water solubility</b>                       | Miscible in water        |                          |
| <b>Solubility(ies)</b>                        | No data available        | None known               |
| <b>Partition coefficient</b>                  | No data available        | None known               |
| <b>Vapour pressure</b>                        | No data available        | None known               |
| <b>Relative density</b>                       | No data available        | None known               |
| <b>Bulk density</b>                           | No data available        |                          |
| <b>Liquid Density</b>                         | No data available        |                          |
| <b>Vapour density</b>                         | No data available        | None known               |
| <b>Particle characteristics</b>               |                          |                          |
| <b>Particle Size</b>                          | No information available |                          |
| <b>Particle Size Distribution</b>             | No information available |                          |

### 9.2. Other information

#### 9.2.1. Information with regard to physical hazard classes

Not applicable

#### 9.2.2. Other safety characteristics

No information available

## SECTION 10: Stability and reactivity

### 10.1. Reactivity

**Reactivity** No information available.

### 10.2. Chemical stability

**Stability** Stable under normal conditions.

#### Explosion data

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** Yes.

**10.3. Possibility of hazardous reactions**

**Possibility of hazardous reactions** None under normal processing.

**10.4. Conditions to avoid**

**Conditions to avoid** Heat, flames and sparks. Exposure to air or moisture over prolonged periods. Excessive heat.

**10.5. Incompatible materials**

**Incompatible materials** Oxidizing agent. Acids. Bases.

**10.6. Hazardous decomposition products**

**Hazardous decomposition products** None known based on information supplied.

**SECTION 11: Toxicological information****11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008****Information on likely routes of exposure****Product Information**

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>   | Specific test data for the substance or mixture is not available. Corrosive by inhalation. (based on components). Inhalation of corrosive fumes/gases may cause coughing, choking, headache, dizziness, and weakness for several hours. Pulmonary edema may occur with tightness in the chest, shortness of breath, bluish skin, decreased blood pressure, and increased heart rate. Inhaled corrosive substances can lead to a toxic edema of the lungs. Pulmonary edema can be fatal. Harmful by inhalation.            |
| <b>Eye contact</b>  | Specific test data for the substance or mixture is not available. Causes serious eye damage. (based on components). Corrosive to the eyes and may cause severe damage including blindness. May cause irreversible damage to eyes.                                                                                                                                                                                                                                                                                         |
| <b>Skin contact</b> | Specific test data for the substance or mixture is not available. Corrosive. (based on components). Causes burns. Toxic in contact with skin.                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Ingestion</b>    | Specific test data for the substance or mixture is not available. Causes burns. (based on components). Ingestion causes burns of the upper digestive and respiratory tracts. May cause severe burning pain in the mouth and stomach with vomiting and diarrhea of dark blood. Blood pressure may decrease. Brownish or yellowish stains may be seen around the mouth. Swelling of the throat may cause shortness of breath and choking. May cause lung damage if swallowed. May be fatal if swallowed and enters airways. |

**Symptoms related to the physical, chemical and toxicological characteristics**

**Symptoms** Redness. Burning. May cause blindness. Coughing and/ or wheezing.

**Acute toxicity****Numerical measures of toxicity**

The following values are calculated based on chapter 3.1 of the GHS document

|                                      |              |
|--------------------------------------|--------------|
| <b>ATEmix (oral)</b>                 | 288.60 mg/kg |
| <b>ATEmix (dermal)</b>               | 799.40 mg/kg |
| <b>ATEmix (inhalation-dust/mist)</b> | 1.44 mg/l    |
| <b>ATEmix (inhalation-vapor)</b>     | 107.70 mg/l  |

**Component Information**

| Chemical name | Oral LD50 | Dermal LD50 | Inhalation LC50 |
|---------------|-----------|-------------|-----------------|
|---------------|-----------|-------------|-----------------|

|             |                      |                          |                         |
|-------------|----------------------|--------------------------|-------------------------|
| Methanol    | = 6200 mg/kg ( Rat ) | = 15840 mg/kg ( Rabbit ) | = 22500 ppm ( Rat ) 8 h |
| Acetic acid | = 3310 mg/kg ( Rat ) | = 1060 mg/kg ( Rabbit )  | = 11.4 mg/L ( Rat ) 4 h |

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                                          |                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin corrosion/irritation</b>         | Classification based on data available for ingredients.                                                                                                                                                                                                                                                                                             |
| <b>Serious eye damage/eye irritation</b> | Classification based on data available for ingredients. Causes burns.                                                                                                                                                                                                                                                                               |
| <b>Respiratory or skin sensitization</b> | No information available.                                                                                                                                                                                                                                                                                                                           |
| <b>Germ cell mutagenicity</b>            | No information available.                                                                                                                                                                                                                                                                                                                           |
| <b>Carcinogenicity</b>                   | No information available.                                                                                                                                                                                                                                                                                                                           |
| <b>Reproductive toxicity</b>             | No information available.                                                                                                                                                                                                                                                                                                                           |
| <b>STOT - single exposure</b>            | Based on the classification criteria of the Globally Harmonized System as adopted in the country or region with which this safety data sheet complies, this product has been determined to cause systemic target organ toxicity from acute exposure. (STOT SE). Causes damage to organs if swallowed. Causes damage to organs in contact with skin. |
| <b>STOT - repeated exposure</b>          | No information available.                                                                                                                                                                                                                                                                                                                           |
| <b>Aspiration hazard</b>                 | No information available.                                                                                                                                                                                                                                                                                                                           |

**11.2. Information on other hazards****11.2.1. Endocrine disrupting properties**

**Endocrine disrupting properties** No information available.

**11.2.2. Other information**

**Other adverse effects** No information available.

|                                           |
|-------------------------------------------|
| <b>SECTION 12: Ecological information</b> |
|-------------------------------------------|

**12.1. Toxicity**

|                                 |                                                                             |
|---------------------------------|-----------------------------------------------------------------------------|
| <b>Ecotoxicity</b>              | Harmful to aquatic life.                                                    |
| <b>Unknown aquatic toxicity</b> | Contains 0 % of components with unknown hazards to the aquatic environment. |

| Chemical name | Algae/aquatic plants | Fish                                                                                                                          | Toxicity to microorganisms | Crustacea |
|---------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------|----------------------------|-----------|
| Methanol      | -                    | LC50: 13500 - 17600mg/L (96h, Lepomis macrochirus)<br>LC50: 18 - 20mL/L (96h, Oncorhynchus mykiss)<br>LC50: 19500 - 20700mg/L | -                          | -         |

|             |   |                                                                                                                        |   |                                    |
|-------------|---|------------------------------------------------------------------------------------------------------------------------|---|------------------------------------|
|             |   | (96h, Oncorhynchus mykiss)<br>LC50: =28200mg/L (96h, Pimephales promelas)<br>LC50: >100mg/L (96h, Pimephales promelas) |   |                                    |
| Acetic acid | - | LC50: =75mg/L (96h, Lepomis macrochirus)<br>LC50: =79mg/L (96h, Pimephales promelas)                                   | - | EC50: =65mg/L (48h, Daphnia magna) |

**12.2. Persistence and degradability**

**Persistence and degradability** No information available.

**12.3. Bioaccumulative potential****Bioaccumulation****Component Information**

| Chemical name | Partition coefficient |
|---------------|-----------------------|
| Methanol      | -0.77                 |
| Acetic acid   | -0.17                 |

**12.4. Mobility in soil**

**Mobility in soil** No information available.

**12.5. Results of PBT and vPvB assessment****PBT and vPvB assessment**

| Chemical name | PBT and vPvB assessment                                                                                                        |
|---------------|--------------------------------------------------------------------------------------------------------------------------------|
| Methanol      | The substance is not PBT / vPvB PBT assessment does not apply Further information relevant for the PBT assessment is necessary |
| Acetic acid   | The substance is not PBT / vPvB PBT assessment does not apply                                                                  |

**12.6. Endocrine disrupting properties**

**Endocrine disrupting properties** No information available.

**12.7. Other adverse effects**

No information available.

## SECTION 13: Disposal considerations

**13.1. Waste treatment methods****Waste from residues/unused products**

Should not be released into the environment. Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging**

Empty containers pose a potential fire and explosion hazard. Do not cut, puncture or weld containers.

## SECTION 14: Transport information

**IATA**

|                                    |                                                                                         |
|------------------------------------|-----------------------------------------------------------------------------------------|
| 14.1 UN number or ID number        | UN2924                                                                                  |
| 14.2 UN proper shipping name       | Flammable liquid, corrosive, n.o.s (Methanol Solution, Acetic acid)                     |
| 14.3 Transport hazard class(es)    | 3                                                                                       |
| Subsidiary hazard class            | 8                                                                                       |
| 14.4 Packing group                 | III                                                                                     |
| Description                        | UN2924, Flammable liquid, corrosive, n.o.s (Methanol Solution, Acetic acid), 3 (8), III |
| 14.5 Environmental hazards         | Not applicable                                                                          |
| 14.6 Special Precautions for Users |                                                                                         |
| Special Provisions                 | A3, A803                                                                                |

**IMDG**

|                                                              |                                                                                                       |
|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 14.1 UN number or ID number                                  | UN2924                                                                                                |
| 14.2 UN proper shipping name                                 | FLAMMABLE LIQUID, CORROSIVE, N.O.S. (Methanol Solution, Acetic acid)                                  |
| 14.3 Transport hazard class(es)                              | 3                                                                                                     |
| Subsidiary hazard class                                      | 8                                                                                                     |
| 14.4 Packing group                                           | III                                                                                                   |
| Description                                                  | UN2924, FLAMMABLE LIQUID, CORROSIVE, N.O.S. (Methanol Solution, Acetic acid), 3 (8), III, (35°C C.C.) |
| 14.5 Environmental hazards                                   | Not applicable                                                                                        |
| 14.6 Special Precautions for Users                           |                                                                                                       |
| Special Provisions                                           | 223, 274                                                                                              |
| EmS-No                                                       | F-E, S-C                                                                                              |
| 14.7 Maritime transport in bulk according to IMO instruments | No information available                                                                              |

**RID**

|                                    |                                                                                          |
|------------------------------------|------------------------------------------------------------------------------------------|
| 14.1 UN number                     | UN2924                                                                                   |
| 14.2 UN proper shipping name       | FLAMMABLE LIQUID, CORROSIVE, N.O.S. (Methanol Solution, Acetic acid)                     |
| 14.3 Transport hazard class(es)    | 3                                                                                        |
| Subsidiary hazard class            | 8                                                                                        |
| 14.4 Packing group                 | III                                                                                      |
| Description                        | UN2924, FLAMMABLE LIQUID, CORROSIVE, N.O.S. (Methanol Solution, Acetic acid), 3 (8), III |
| 14.5 Environmental hazards         | Not applicable                                                                           |
| 14.6 Special Precautions for Users |                                                                                          |
| Special Provisions                 | 274                                                                                      |
| Classification code                | FC                                                                                       |

**ADR**

|                                    |                                                                                        |
|------------------------------------|----------------------------------------------------------------------------------------|
| 14.1 UN number or ID number        | 2924                                                                                   |
| 14.2 UN proper shipping name       | FLAMMABLE LIQUID, CORROSIVE, N.O.S. (Methanol Solution, Acetic acid)                   |
| 14.3 Transport hazard class(es)    | 3                                                                                      |
| Subsidiary class                   | 8                                                                                      |
| 14.4 Packing group                 | III                                                                                    |
| Description                        | 2924, FLAMMABLE LIQUID, CORROSIVE, N.O.S. (Methanol Solution, Acetic acid), 3 (8), III |
| 14.5 Environmental hazards         | Not applicable                                                                         |
| 14.6 Special Precautions for Users |                                                                                        |
| Special Provisions                 | 274                                                                                    |
| Classification code                | FC                                                                                     |
| Tunnel restriction code            | (D/E)                                                                                  |

**SECTION 15: Regulatory information****15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture****National regulations****France****Occupational Illnesses (R-463-3, France)**

| Chemical name       | French RG number | Title |
|---------------------|------------------|-------|
| Methanol<br>67-56-1 | RG 84            | -     |

**Germany**

**Water hazard class (WGK)** obviously hazardous to water (WGK 2)

**European Union**

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work.

**Authorizations and/or restrictions on use:**

This product contains one or more substance(s) subject to restriction (Regulation (EC) No. 1907/2006 (REACH), Annex XVII)

| Chemical name         | Restricted substance per REACH Annex XVII | Substance subject to authorization per REACH Annex XIV |
|-----------------------|-------------------------------------------|--------------------------------------------------------|
| Methanol - 67-56-1    | 69.<br>75.                                | -                                                      |
| Acetic acid - 64-19-7 | 75.                                       | -                                                      |

**Persistent Organic Pollutants**

Not applicable

**Dangerous substance category per Seveso Directive (2012/18/EU)**

H3 - STOT SPECIFIC TARGET ORGAN TOXICITY - SINGLE EXPOSURE

P5a - FLAMMABLE LIQUIDS

P5b - FLAMMABLE LIQUIDS

P5c - FLAMMABLE LIQUIDS

**Named dangerous substances per Seveso Directive (2012/18/EU)**

| Chemical name      | Lower-tier requirements (tons) | Upper-tier requirements (tons) |
|--------------------|--------------------------------|--------------------------------|
| Methanol - 67-56-1 | 500                            | 5000                           |

**Ozone-depleting substances (ODS) regulation (EC) 1005/2009**

Not applicable

**Plant protection products directive (91/414/EEC)**

| Chemical name         | Plant protection products directive (91/414/EEC) |
|-----------------------|--------------------------------------------------|
| Acetic acid - 64-19-7 | Plant protection agent                           |

**EU - Biocides****International Inventories**

Contact supplier for inventory compliance status

**15.2. Chemical safety assessment****Chemical Safety Report**

No information available

## SECTION 16: Other information

**Key or legend to abbreviations and acronyms used in the safety data sheet****Full text of H-Statements referred to under section 3**

H225 - Highly flammable liquid and vapor

H226 - Flammable liquid and vapor

H301 - Toxic if swallowed

H311 - Toxic in contact with skin

H314 - Causes severe skin burns and eye damage

H331 - Toxic if inhaled

H370 - Causes damage to organs

**Legend**

SVHC: Substances of Very High Concern for Authorisation:

**Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

TWA TWA (time-weighted average) STEL STEL (Short Term Exposure Limit)  
 Ceiling Maximum limit value \* Skin designation

| Classification procedure                                        |                       |
|-----------------------------------------------------------------|-----------------------|
| Classification according to Regulation (EC) No. 1272/2008 [CLP] | Method Used           |
| Acute oral toxicity                                             | Calculation method    |
| Acute dermal toxicity                                           | Calculation method    |
| Acute inhalation toxicity - gas                                 | Calculation method    |
| Acute inhalation toxicity - Vapour                              | Calculation method    |
| Acute inhalation toxicity - dust/mist                           | Calculation method    |
| Skin corrosion/irritation                                       | Calculation method    |
| Serious eye damage/eye irritation                               | Calculation method    |
| Respiratory sensitisation                                       | Calculation method    |
| Skin sensitisation                                              | Calculation method    |
| Mutagenicity                                                    | Calculation method    |
| Carcinogenicity                                                 | Calculation method    |
| Reproductive toxicity                                           | Calculation method    |
| STOT - single exposure                                          | Calculation method    |
| STOT - repeated exposure                                        | Calculation method    |
| Acute aquatic toxicity                                          | Calculation method    |
| Chronic aquatic toxicity                                        | Calculation method    |
| Aspiration hazard                                               | Calculation method    |
| Ozone                                                           | Calculation method    |
| Corrosive to metals                                             | On basis of test data |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 European Chemicals Agency (ECHA) Committee for Risk Assessment (ECHA\_RAC)  
 European Chemicals Agency (ECHA) (ECHA\_API)  
 EPA (Environmental Protection Agency)  
 Acute Exposure Guideline Level(s) (AEGL(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
 Organization for Economic Co-operation and Development Screening Information Data Set  
 World Health Organization

**Revision Note** Reformatted and updated existing information

**Revision date** 31-Aug-2022

**This material safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006**

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

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**End of Safety Data Sheet**